

Some fun with differential equations

Let's consider the following equation, and how to solve it as a series expansion using Mathematica,

$$P(x) \frac{d^2 y(x)}{dx^2} + Q(x) \frac{dy(x)}{dx} + R(x)y(x) = 0,$$

for some P, Q, R, let's say

$$P(x) = x^2,$$

$$Q(x) = 0,$$

$$R(x) = \cos(x),$$

just to give some example. And for the boundary conditions,

$$y(0) = 1,$$

$$y'(0) = 0.$$

Defining the problem

First, let's define the equations, to define a function of some input argument:

- Square brackets around the argument
- Use an underscore after the argument
- “:=” is for defining a function which only evaluates when called whereas “=” is for assigning a value at the time of the definition.

I've done one definition below, you can fill out the others.

```
In[ ]:= ClearAll[P, Q, R]
```

```
P[x_] := (1 + x^2)
```

```
Q[x_] := 0
```

```
R[x_] := Cos[x]
```

Hopefully the following explains what is going on with the different assignments.

```
In[ ]:= z = 1;
```

```
testEquals[z_] = z^2
```

```
testColonEquals[z_] := z^2;
```

```
Out[ ]= 1
```

```
In[ ]:= testEquals[2]
```

```
testColonEquals[2]
```

```
Out[ ]= 1
```

```
Out[ ]= 4
```

In case we want to use the letter z below, we will clear the value that we have assigned to it.

```
In[*]:= ClearAll[z]
```

Seeing if Mathematica knows the answer

Mathematica knows lots of differential equations, so sometimes it just knows the answer.

Chebyshev equation

Below I show that it knows the answer to the Chebyshev equation (with $\alpha=1$), though it might express them in a funny way. To test this we use “DSolve”. Hover over it and click to see the documentation.

Note the use of “==” here, which is not assignment but mathematical equality.

```
In[*]:= ySolutions = DSolve[
```

```
(1 - x^2) y''[x] - x y'[x] + y[x] == 0,
```

```
y[x],
```

```
x
```

```
]
```

```
Out[*]:= {{y[x] -> c1 Cosh[ $\frac{2 \sqrt{1-x^2} \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{-1+x^2}}$ ] - i c2 Sinh[ $\frac{2 \sqrt{1-x^2} \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{-1+x^2}}$ ]]}}
```

Curly brackets here mean a list. That there are two layers means we have a list of a list, though only of length 1. To get an element of a list, use double square brackets. Note also that Mathematica starts counting from 1, not 0.

```
In[*]:= {the, cat, in, the, hat}[[2]]
```

```
Out[*]:= cat
```

There is some more notation here in the solution above. The arrow is a kind of find and replace. To utilise it, use the notation “\.” as follows.

```
In[*]:= ySolution[x_] = y[x] /. ySolutions[[1]]
```

```
Out[*]:= c1 Cosh[ $\frac{2 \sqrt{1-x^2} \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{-1+x^2}}$ ] - i c2 Sinh[ $\frac{2 \sqrt{1-x^2} \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{-1+x^2}}$ ]]
```

Let’s test that this really is a solution, by putting the solution back into the equation. Note that I am defining a variable with [1] after it. This is useful if I want to manipulate the object and define further versions of it below.

```
In[*]:= shouldBeZero[1] = ((1 - x^2) y''[x] - x y'[x] + y[x]) /. ySolutions[[1]]
```

```
Out[*]:= c1 Cosh[ $\frac{2 \sqrt{1-x^2} \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{-1+x^2}}$ ] - i c2 Sinh[ $\frac{2 \sqrt{1-x^2} \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{-1+x^2}}$ ]] - x y'[x] + (1 - x^2) y''[x]
```

That didn’t work! Why not? When something doesn’t work in Mathematica, a good trick is to see either

the InputForm (or the FullForm) of the output.

```
In[*]:= InputForm[shouldBeZero[1]]
Out[*]//InputForm=
C[1]*Cosh[(2*Sqrt[1 - x^2]*ArcTan[Sqrt[1 - x^2]/(1 + x)])/Sqrt[-1 + x^2]] - I*C[2]*Sinh[(2*Sqrt[1 -
(1 - x^2)*Derivative[2][y][x]
```

```
In[*]:= InputForm[ySolutions[[1]]]
Out[*]//InputForm=
{y[x] -> C[1]*Cosh[(2*Sqrt[1 - x^2]*ArcTan[Sqrt[1 - x^2]/(1 + x)])/Sqrt[-1 + x^2]] - I*C[2]*Sinh[(2*
```

You can see that while $y[x]$ was replaced, $\text{Derivative}[1][y][x]$ wasn't. Mathematica isn't smart like this.

The find and replace methods will only replace exact matches.

```
In[*]:= shouldBeZero[2] =
shouldBeZero[1] /. {y'[x] -> D[ySolution[x], x], y''[x] -> D[ySolution[x], x, x]}
```

$$\begin{aligned} \text{Out[*]} = & c_1 \cosh\left[\frac{2\sqrt{1-x^2} \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{-1+x^2}}\right] - i c_2 \sinh\left[\frac{2\sqrt{1-x^2} \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{-1+x^2}}\right] - \\ & x \left(-i \left(\frac{2\sqrt{1-x^2} \left(-\frac{x}{(1+x)\sqrt{1-x^2}} - \frac{\sqrt{1-x^2}}{(1+x)^2} \right)}{\sqrt{-1+x^2} \left(1 + \frac{1-x^2}{(1+x)^2} \right)} - \frac{2x\sqrt{1-x^2} \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{(-1+x^2)^{3/2}} - \frac{2x \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{1-x^2} \sqrt{-1+x^2}} \right) \right. \\ & \left. c_2 \cosh\left[\frac{2\sqrt{1-x^2} \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{-1+x^2}}\right] + \right. \\ & \left(\frac{2\sqrt{1-x^2} \left(-\frac{x}{(1+x)\sqrt{1-x^2}} - \frac{\sqrt{1-x^2}}{(1+x)^2} \right)}{\sqrt{-1+x^2} \left(1 + \frac{1-x^2}{(1+x)^2} \right)} - \frac{2x\sqrt{1-x^2} \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{(-1+x^2)^{3/2}} - \frac{2x \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{1-x^2} \sqrt{-1+x^2}} \right) \\ & \left. c_1 \sinh\left[\frac{2\sqrt{1-x^2} \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{-1+x^2}}\right] \right) + \\ & (1-x^2) \left(\left(\frac{2\sqrt{1-x^2} \left(-\frac{x}{(1+x)\sqrt{1-x^2}} - \frac{\sqrt{1-x^2}}{(1+x)^2} \right)}{\sqrt{-1+x^2} \left(1 + \frac{1-x^2}{(1+x)^2} \right)} - \frac{2x\sqrt{1-x^2} \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{(-1+x^2)^{3/2}} - \frac{2x \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{1-x^2} \sqrt{-1+x^2}} \right)^2 \right. \\ & \left. c_1 \cosh\left[\frac{2\sqrt{1-x^2} \operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{-1+x^2}}\right] - i \left(-\frac{2\sqrt{1-x^2} \left(-\frac{x}{(1+x)\sqrt{1-x^2}} - \frac{\sqrt{1-x^2}}{(1+x)^2} \right) \left(-\frac{2x}{(1+x)^2} - \frac{2(1-x^2)}{(1+x)^3} \right)}{\sqrt{-1+x^2} \left(1 + \frac{1-x^2}{(1+x)^2} \right)^2} + \right. \right. \\ & \left. \frac{2\sqrt{1-x^2} \left(-\frac{x^2}{(1+x)(1-x^2)^{3/2}} + \frac{2x}{(1+x)^2 \sqrt{1-x^2}} - \frac{1}{(1+x)\sqrt{1-x^2}} + \frac{2\sqrt{1-x^2}}{(1+x)^3} \right)}{\sqrt{-1+x^2} \left(1 + \frac{1-x^2}{(1+x)^2} \right)} - \right. \end{aligned}$$

$$\begin{aligned}
& \frac{4x\sqrt{1-x^2}\left(-\frac{x}{(1+x)\sqrt{1-x^2}} - \frac{\sqrt{1-x^2}}{(1+x)^2}\right)}{(-1+x^2)^{3/2}\left(1+\frac{1-x^2}{(1+x)^2}\right)} - \frac{4x\left(-\frac{x}{(1+x)\sqrt{1-x^2}} - \frac{\sqrt{1-x^2}}{(1+x)^2}\right)}{\sqrt{1-x^2}\sqrt{-1+x^2}\left(1+\frac{1-x^2}{(1+x)^2}\right)} + \\
& \frac{6x^2\sqrt{1-x^2}\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{(-1+x^2)^{5/2}} + \frac{4x^2\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{1-x^2}(-1+x^2)^{3/2}} - \\
& \left. \frac{2\sqrt{1-x^2}\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{(-1+x^2)^{3/2}} - \frac{2x^2\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{(1-x^2)^{3/2}\sqrt{-1+x^2}} - \frac{2\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{1-x^2}\sqrt{-1+x^2}} \right) \mathbf{c}_2 \\
& \operatorname{Cosh}\left[\frac{2\sqrt{1-x^2}\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{-1+x^2}}\right] + \left(-\frac{2\sqrt{1-x^2}\left(-\frac{x}{(1+x)\sqrt{1-x^2}} - \frac{\sqrt{1-x^2}}{(1+x)^2}\right)\left(-\frac{2x}{(1+x)^2} - \frac{2(1-x^2)}{(1+x)^3}\right)}{\sqrt{-1+x^2}\left(1+\frac{1-x^2}{(1+x)^2}\right)^2} + \right. \\
& \left. \frac{2\sqrt{1-x^2}\left(-\frac{x^2}{(1+x)(1-x^2)^{3/2}} + \frac{2x}{(1+x)^2\sqrt{1-x^2}} - \frac{1}{(1+x)\sqrt{1-x^2}} + \frac{2\sqrt{1-x^2}}{(1+x)^3}\right)}{\sqrt{-1+x^2}\left(1+\frac{1-x^2}{(1+x)^2}\right)} - \right. \\
& \frac{4x\sqrt{1-x^2}\left(-\frac{x}{(1+x)\sqrt{1-x^2}} - \frac{\sqrt{1-x^2}}{(1+x)^2}\right)}{(-1+x^2)^{3/2}\left(1+\frac{1-x^2}{(1+x)^2}\right)} - \frac{4x\left(-\frac{x}{(1+x)\sqrt{1-x^2}} - \frac{\sqrt{1-x^2}}{(1+x)^2}\right)}{\sqrt{1-x^2}\sqrt{-1+x^2}\left(1+\frac{1-x^2}{(1+x)^2}\right)} + \\
& \frac{6x^2\sqrt{1-x^2}\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{(-1+x^2)^{5/2}} + \frac{4x^2\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{1-x^2}(-1+x^2)^{3/2}} - \frac{2\sqrt{1-x^2}\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{(-1+x^2)^{3/2}} - \\
& \left. \frac{2x^2\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{(1-x^2)^{3/2}\sqrt{-1+x^2}} - \frac{2\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{1-x^2}\sqrt{-1+x^2}} \right) \mathbf{c}_1 \operatorname{Sinh}\left[\frac{2\sqrt{1-x^2}\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{-1+x^2}}\right] - \\
& i \left(\frac{2\sqrt{1-x^2}\left(-\frac{x}{(1+x)\sqrt{1-x^2}} - \frac{\sqrt{1-x^2}}{(1+x)^2}\right)}{\sqrt{-1+x^2}\left(1+\frac{1-x^2}{(1+x)^2}\right)} - \frac{2x\sqrt{1-x^2}\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{(-1+x^2)^{3/2}} - \frac{2x\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{1-x^2}\sqrt{-1+x^2}} \right)^2 \\
& \left. \mathbf{c}_2 \operatorname{Sinh}\left[\frac{2\sqrt{1-x^2}\operatorname{ArcTan}\left[\frac{\sqrt{1-x^2}}{1+x}\right]}{\sqrt{-1+x^2}}\right] \right)
\end{aligned}$$

Still not zero! Let's see what FullSimplify can do.

```
In[ ]:= FullSimplify[shouldBeZero[2]]
```

```
Out[ ]:= 0
```

Our actual equation

Okay, test the problem we actually want to solve.

Series solution

Okay, so if you have got here, you have realised that Mathematica doesn't just know the answer.

Let's define a series solution ansatz for $y(x)$.

```
In[ ]:= ySeries[x_, n_] := Sum[ya[a] x^a, {a, 0, n}];
```

```
In[ ]:= ySeries[x, 4]
```

```
Out[ ]:= ya[0] + x ya[1] + x^2 ya[2] + x^3 ya[3] + x^4 ya[4]
```

Now let's apply Mathematica's Series function. For documentation, hover over the function and click.

```
In[ ]:= n = 4;
```

```
eqnExpansion = Series[
  P[x] × D[ySeries[x, n], x, x] +
  Q[x] × D[ySeries[x, n], x] +
  R[x] × ySeries[x, n],
  {x, 0, n}
]
```

```
Clear[n]
```

```
Out[ ]:= (ya[0] + 2 ya[2]) + (ya[1] + 6 ya[3]) x +  $\left(-\frac{ya[0]}{2} + 3 ya[2] + 12 ya[4]\right) x^2 +$ 
 $\left(-\frac{ya[1]}{2} + 7 ya[3]\right) x^3 + \frac{1}{24} (ya[0] - 12 ya[2] + 312 ya[4]) x^4 + O[x]^5$ 
```

Let's define a function to get the nth order term.

```
In[ ]:= eqnN[n_] := SeriesCoefficient[
  P[x] × D[ySeries[x, n], x, x] +
  Q[x] × D[ySeries[x, n], x] +
  R[x] × ySeries[x, n],
  {x, 0, n}
]
```

```
eqnN[1]
```

```
Out[ ]:= ya[1]
```

Now we want to solve the first few equations.

```
In[ ]:= results = Table[If[i == 0, 1, 0], {i, 0, 4}]
yApproxSol = Solve[
  Thread[results == Table[eqnN[n], {n, 0, 4}]]
  , Table[ya[i], {i, 0, 4}]
]
Out[ ]:= {1, 0, 0, 0, 0}
Out[ ]:= {{ya[0] → 1, ya[1] → 0, ya[2] →  $\frac{1}{6}$ , ya[3] → 0, ya[4] →  $\frac{1}{312}$ }}
```

```
In[ ]:= yApprox[n_ /; n ≤ 4] := Sum[ya[i] xi, {i, 0, n}] /. yApproxSol[[1]]
```

Numerical solution

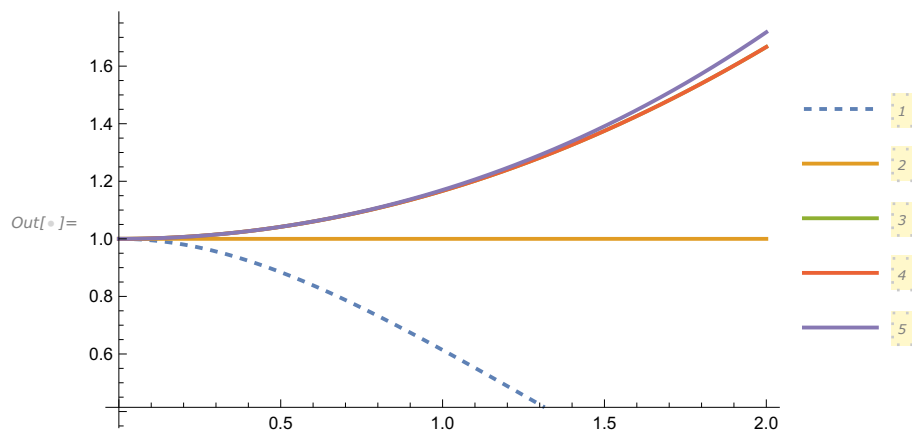
For numerical calculations Mathematica typically defines functions beginning with “N”, like NDSolve.

```
In[ ]:= ySolutionN = NDSolve[
  {P[x] × y'[x] + Q[x] × y'[x] + R[x] × y[x] == 0, y[0] == 1, y'[0] == 0},
  y[x],
  {x, 0, 2}
]
Out[ ]:= {{y[x] → InterpolatingFunction[ Domain: {{0., 2.}}
Output: scalar][x]}}
```

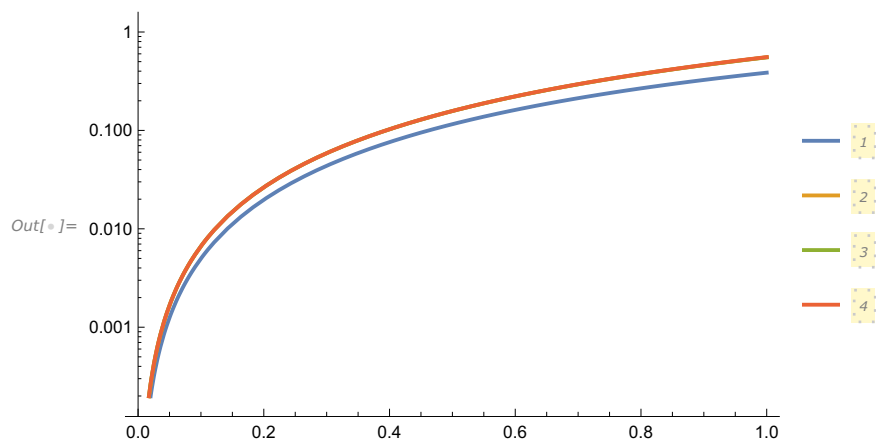
```
In[ ]:=
```

Making some plots

```
In[ ]:= Plot[
  {y[x] /. ySolutionN[1],
   yApprox[1],
   yApprox[2],
   yApprox[3],
   yApprox[4]}
  , {x, 0, 2}, PlotLegends → Automatic,
  PlotStyle → {Dashed, Automatic, Automatic, Automatic, Automatic}]
```



```
In[ ]:= LogPlot[
  {Abs[yApprox[1] - y[x] /. ySolutionN[1]],
   Abs[yApprox[2] - y[x] /. ySolutionN[1]],
   Abs[yApprox[3] - y[x] /. ySolutionN[1]],
   Abs[yApprox[4] - y[x] /. ySolutionN[1]]}
  , {x, 10-10, 1}, PlotLegends → Automatic]
```



Saving the data

